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The University of Texas at Austin

Lmod: The Lua-Based Environment Module System

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What is Lmod?

- Lmod is a **Lua-based environment module system** for managing software stacks on HPC systems.
- Supports both **Lua and Tcl modulefiles**.
- Enables **hierarchical MODULEPATHs**, **module families**, and **site policy enforcement**.
- Used at major HPC centers worldwide including TACC, NERSC, CSCS, and BSC.

Recap: Where We Left Off (SC24)

- Focused on improving modulefile scripting, hooks, and hierarchy handling.
- Introduced the next-generation tracking database for usage collection.
- Simplified configuration and policy definition for large sites.
- Feedback from SC24 deployments drove the Lmod 9 feature roadmap.

From 8.x to 9.x – What Changed Under the Hood

- Consolidated over 60 feature and performance patches into the new 9.x branch.
- New dependency engine reduces redundant file reads and improves load order resolution.
- Tracking database re-engineered – up to **100× smaller footprint** at TACC.
- Expanded site-policy features: irreversible unloads, hide/forbid logic.
- Maintains full backward compatibility with existing 8.x stacks.

New Features and Enhancements

- **Irreversible mode**: unloading can now set environment variables for cleanup.
- **depends_on_any()**: allows dependency on any member of a module set.
- **Enhanced hide{}/forbid{}**: expression-based control by user, group, or date.
- **Optional tracking v2**: database shrinks 100× while preserving analytics.
- **New hooks (e.g., decorate_module)** for module tagging and logging.

Performance and Reliability

- Spider cache and dependency engine now reduce ‘module avail‘ from 4s ⟶ <0.5s on 40k-module trees.
- Refactored collection handling avoids NFS metadata bottlenecks.
- Improved `family()` logic speeds hierarchical resolution.
- Expanded compatibility: bash, zsh, fish, and nushell all supported.

Documentation and Usability

- Major expansion of lmod.readthedocs.io with in-depth technical docs.
- Added sections on advanced hooks, tracking v2, and policy examples.
- Clear migration guide from 8.x ⟶ 9.x.
- Real-world stats: TACC reduced usage DB size from 1 GB/day ⟶ 10 MB/day.

Site Policy and Governance

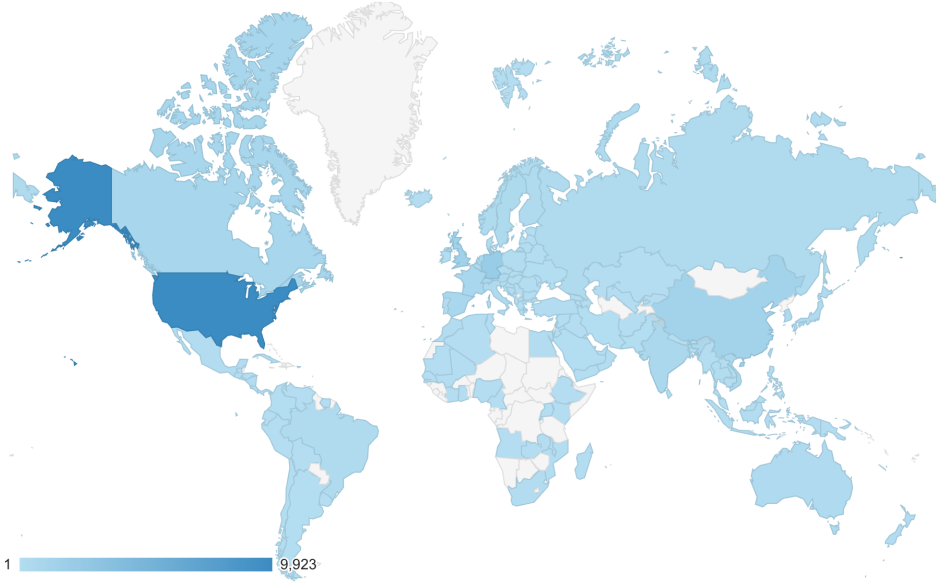
- Conditional **hide()/forbid()** rules by user, group, or time window.
- Deprecation warnings for outdated compilers or toolchains.
- Irreversible unloads enable post-cleanup or variable resets.
- Simplifies compliance and software lifecycle management.

Operational Impact at TACC and Partner Sites

- On Frontera and Stampede3: module load latency **reduced by 40%** in deep hierarchies.
- Tracking database ingest reduced **100×**.
- GPU/CPU module tagging integrated via site hooks for usage analytics.
- Single `lmod_config.lua` replaced multiple legacy init scripts.

Looking Ahead

- Ongoing performance profiling for massive module trees (>100k modules).
- Additional shell and scheduler integration (Slurm, PBSPro).
- Exploring distributed tracking and federated analytics.
- Continued collaboration via GitHub and EUM workshops.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Get Involved

- Source: github.com/TACC/Lmod
- Docs: lmod.readthedocs.io
- Mailing lists: `lmod-announce`, `lmod-users`.
- Contribute: report issues, propose hooks, share use cases.

Thank You

Questions? Feedback?

Contact: `mcawood@tacc.utexas.edu`

`lmod.readthedocs.io`